

6. เงื่อนไขทางคลินิกของการเบิกจ่ายค่ายา Brigatinib/Ceritinib ดังต่อไปนี้

6.1 มีการวินิจฉัยโรคตามหลักเกณฑ์ในข้อ 4.1 และ 4.2

6.2 ใช้เป็นยาขนานแรก หรือขนานที่ 2 ในผู้ป่วยมีการกำเริบของโรค (progressive disease) หลังได้ยาเคมีบำบัด

6.3 ผู้ป่วยต้องอยู่ในสภาพที่แข็งแรง ECOG 0 - 2 กรณีที่เป็น ECOG 3 ต้องเป็นภาวะที่เกิดจากโรคมะเร็งปอด

7. ข้อห้ามของการใช้ยา

ผู้ป่วยที่มี significant comorbidities ที่การรักษาไม่สามารถเพิ่มคุณภาพชีวิต แต่ทำให้เกิดผลข้างเคียงจากยา ผู้ป่วยที่ไม่สามารถสื่อสารเพื่อบอกอาการโรคหรืออาการข้างเคียงที่เกิดจากการรักษาได้

8. ขนาดยาที่แนะนำ

- Brigatinib 90 มก. วันละ 1 ครั้ง ใน 7 วันแรกของการเริ่มต้นใช้ยา ตามด้วย Brigatinib 180 มก. ต่อวัน ตั้งแต่วันที่ 8 เป็นต้นไป กรณีที่มีการปรับลดยา ให้ใช้วิธีลดจำนวนเม็ดยา Brigatinib 180 มก. ต่อสัปดาห์
- Ceritinib 450 มก. รับประทานพร้อมอาหาร วันละ 1 ครั้ง

หมายเหตุ

1. ปริมาณการเบิกยา ครั้งละ 1 เดือน ใน 3 เดือนแรก และไม่เกินครั้งละ 2 เดือน ในเดือนต่อ ๆ ไป
2. มีให้เบิกค่ายา Brigatinib ความแรง 30 มก.
3. กรณีที่ใช้ยา Brigatinib หลังจากสัปดาห์แรก ให้เบิกได้เฉพาะความแรง 180 มก. เท่านั้น

9. การประเมินผู้ป่วยระหว่างการรักษา

9.1 ประเมินสภาพผู้ป่วยตามมาตรฐานวิชาชีพ ทุก 1 เดือน ในช่วง 3 เดือนแรก ได้แก่ บันทึกอาการ การตรวจร่างกาย ปัญหาของผู้ป่วย และผลข้างเคียงที่เกิดจากการรักษา

9.2 ประเมินผลการรักษาทุก 2 - 3 เดือน ควรได้ผลการรักษาก่อน 3 เดือน เพื่อใช้ในการขออนุมัติการเบิกจ่ายาครั้งต่อไป

10. เกณฑ์การหยุดยา ให้หยุดยา Brigatinib/Ceritinib เมื่อตรวจพบข้อใดข้อหนึ่งดังต่อไปนี้

10.1 ผู้ป่วยไม่ตอบสนองต่อการรักษา เป็น progressive disease ยกเว้นกรณีที่เป็น localized brain metastasis และไม่มี extra cranial progression

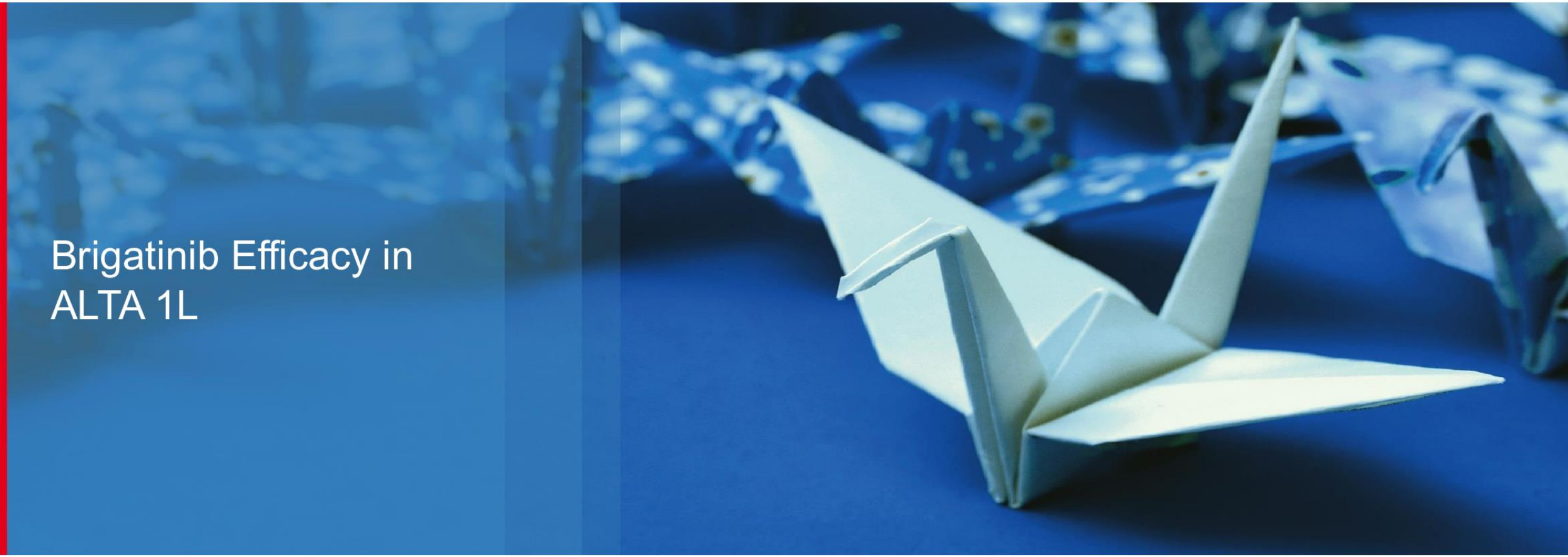
10.2 เกิดผลข้างเคียงจากยา ที่รุนแรงระดับ 3 - 4 ที่ไม่สามารถควบคุมได้

แนวทางกำกับการเบิกจ่ายค่ายา Ceritinib ในโรคมะเร็งปอดระยะแพร่กระจาย ชนิด ALK+

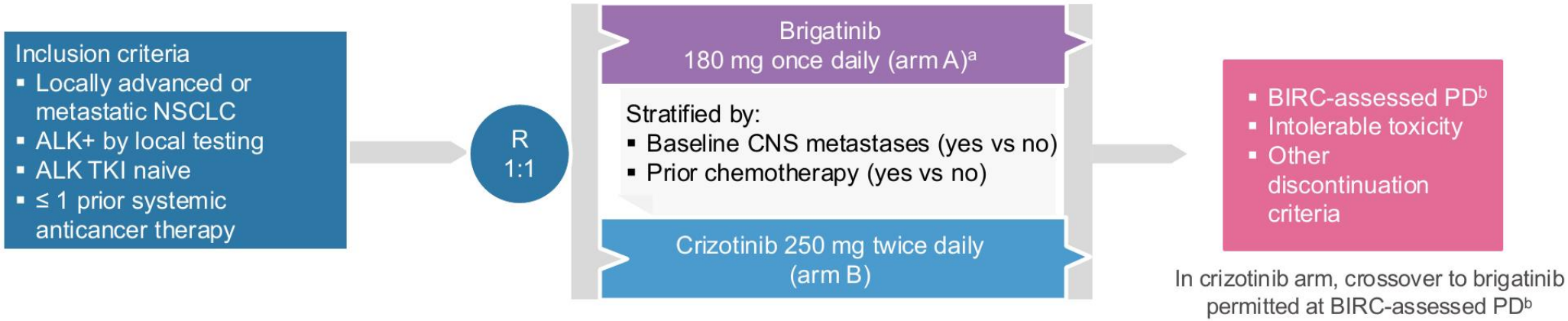
1. **เกณฑ์อนุมัติการเบิกค่ายา Ceritinib ได้แก่**
   - 1.1 ต้องมีผลการตรวจทางพยาธิวิทยาของมะเร็งปอดชนิด Non-small cell lung cancer ร่วมกับมีการตรวจพบความผิดปกติของยีน ALK rearrangement ในเนื้อเยื่อเซลล์มะเร็ง
   - 1.2 ใช้เป็นยาขนานแรกในผู้ป่วยโรคมะเร็งปอดชนิด Non-small cell lung cancer ระยะแพร่กระจายหรือกลับเป็นซ้ำ (metastatic or recurrent disease) และมีรอยโรคที่สามารถประเมินได้
   - 1.3 ใช้เป็นยาขนานที่ 2 และ 3 ในผู้ป่วยมีการกำเริบของโรค (progressive disease) หลังได้ยาเคมีบำบัด
   - 1.4 ผู้ป่วยต้องอยู่ในสภาพที่แข็งแรง ECOG 0 - 2 กรณีที่เป็น ECOG 3 ต้องเป็นภาวะที่เกิดจากโรคมะเร็งปอด
   - 1.5 ไม่อนุมัติการใช้ยากลุ่ม ALK inhibitor ขนานที่ 2 หรือ 3 ในผู้ป่วยที่ไม่ตอบสนองต่อยา ALK inhibitor ขนานแรก ceritinib มาก่อน
   - 1.6 ผู้ป่วยรายเดิม ที่ใช้ยา ALK inhibitor ขนานอื่นอยู่ ให้ลงทะเบียนเพื่อเบิกจ่ายตรงต่อได้ที่ http://mra.or.th/ocpaoldcase/index.php

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Brigatinib Efficacy in ALTA 1L



ALTA-1L: ALK TKI–Naive ALK+ NSCLC Phase 3 Study (NCT02737501)[1-4]



Disease assessment (including brain MRI for all patients) every 8 weeks through cycle 14[c]

Additional key inclusion criteria: ECOG PS 0-2, measurable disease, ≤ 1 prior regimen of systemic therapy in the advanced setting

Primary endpoint: BIRC-assessed PFS per RECIST v1.1[d]

Statistical considerations

- ≈ 270 total patients (198 events), 135 in each arm to detect a 6-month improvement in PFS (HR, 0.625), assuming 10-month PFS in crizotinib arm
- 2 interim analyses and a final analysis were planned for the primary endpoint of BIRC-assessed PFS[d]

[a] 180 mg once daily with 7-day lead-in at 90 mg once daily; [b] Arm B crossover to brigatinib allowed at BIRC-assessed disease progression following a 10-day washout period; [c] Disease was assessed at screening, every 8 weeks through cycle 14 (28 days/cycle), and then every 12 weeks through treatment discontinuation; [d] Statistical significance for the primary endpoint was achieved at the first interim analysis.

1. Camidge DR, et al. *N Engl J Med*. 2018;379:2027-2039; 2. Camidge DR, et al. ESMO Asia 2019 [abstract LBA1]. 3. Camidge DR, et al. *J Thorac Oncol*. 2021;16:2091-2108; 4. Company's data on file (ALTA-1L leadership presentation_FA_topline; 2021).

ALTA-1L: Baseline Characteristics

Characteristic[1-3]		Brigatinib (n = 137)	Crizotinib (n = 138)
Median age (range), years		58 (27-86)	60 (29-89)
Sex, %	Female	50	59
Race, %	White Asian Black Unknown	55 43 0 1	62 36 1 1
ECOG PS, %	0 1 2	39 55 5	38 57 5
Stage of disease at study entry, %	IIIB IV	6 94	9 91
Median time since initial diagnosis (range), months		1.7 (0.1-145.3)	1.5 (0.3-189.8)
ALK status assessed locally by FDA-approved test, %[a,b]	Yes	90	81
Brain metastases at baseline, %[c]	Yes	29	30
Prior RT to the brain, %	Yes	40	39
Prior RT to the brain[b]	WBRT SRS	11 3	9 6
Prior chemotherapy, %[d]	Yes	26	27

Final data cutoff: January 29, 2021; median follow-up: 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm.
[a] Patients with ALK+ status that was confirmed locally by Vysis FISH or Ventana IHC; [b] Data based on the second interim analyses[2]; [c] As assessed by INV; [d] Prior chemotherapy was defined as completion of ≥ 1 full cycle of chemotherapy in the locally advanced or metastatic setting; [e] In crizotinib arm, crossover to brigatinib was permitted at BIRC-assessed PD with a 10-day washout period required.

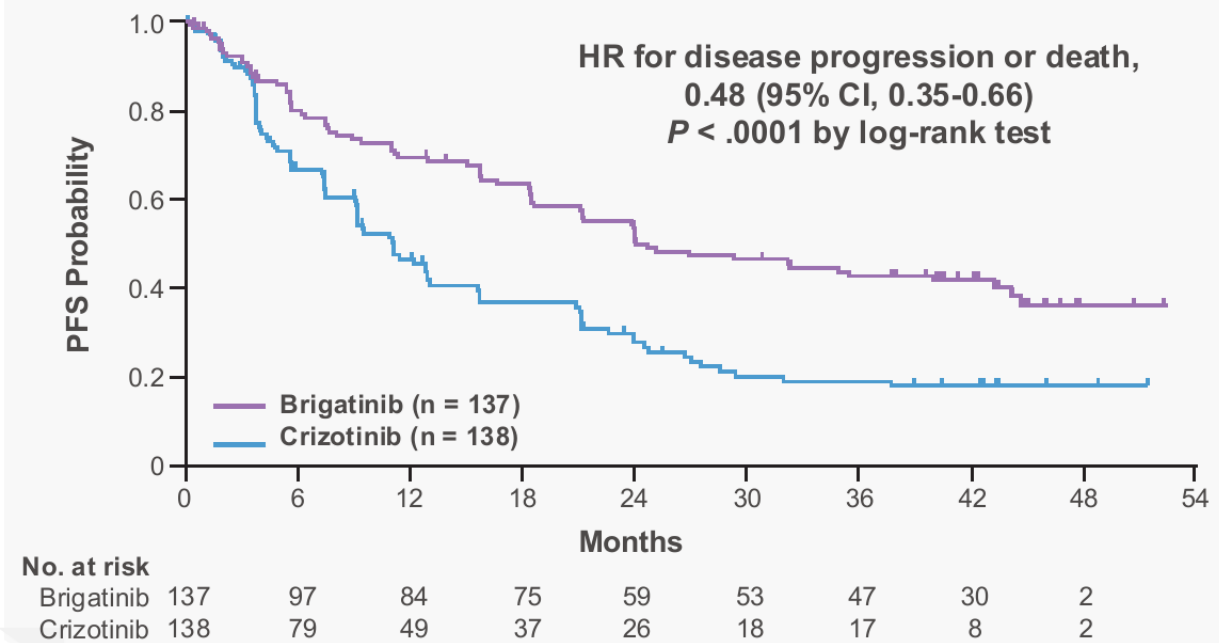
- Median duration of treatment
  - IA1: 9.2 months (range, 0.1-18.4 months) in the brigatinib arm and 7.4 months (range, 0.1-19.2 months) in the crizotinib arm[1]
  - **Final analysis**: 34.9 months (range, 0.1-52.4 months) in the brigatinib arm and 9.3 months (range, 0.1-51.5 months) in the crizotinib arm[3]
- Patient disposition
  - **Final analysis**: all patients had discontinued treatment[3]
    - 58 patients (42%) in the brigatinib arm and 16 patients (12%) in the crizotinib arm were still receiving study drug before the study was terminated
    - 65 patients (47%) discontinued crizotinib and crossed over to receive brigatinib, according to the protocol[3,e]

1. Camidge DR, et al. *N Engl J Med.* 2018;379:2027-2039; 2. Camidge DR, et al. ESMO Asia 2019 [abstract LBA1]; 3. Camidge DR, et al. *J Thorac Oncol.* 2021;16:2091-2108.

ALTA-1L Final Analysis: Updated PFS

- Brigatinib significantly reduced the risk of progression or death by 52% and 57% vs crizotinib by BIRC and INV, respectively
  - With a follow-up of 40 months, brigatinib demonstrated consistent overall efficacy, confirming results seen in the first and second interim analyses

Final Analysis: BIRC-Assessed Systemic PFS (primary endpoint)

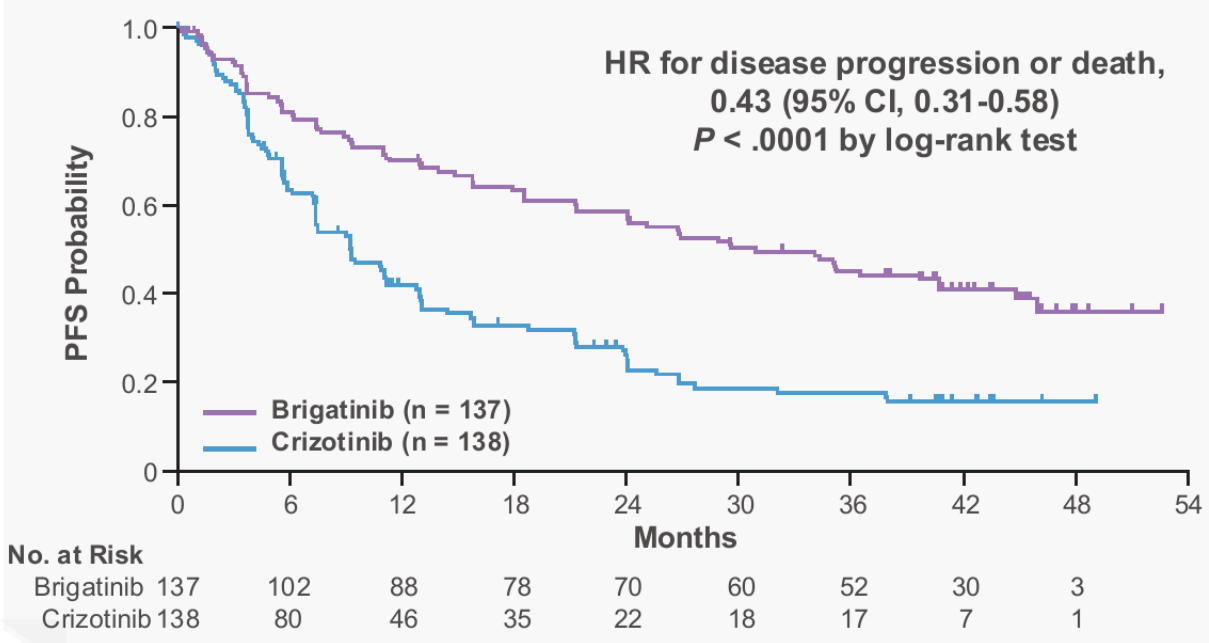


No. at risk

Brigatinib	137	97	84	75	59	53	47	30	2
Crizotinib	138	79	49	37	26	18	17	8	2

BIRC-Assessed PFS	Median PFS (95% CI), months	HR (95% CI)	3-Year PFS Rate (95% CI), %	4-Year PFS Rate (95% CI), %
Brigatinib (n = 137)	24.0 (18.5-43.2)	0.48 (0.35-0.66) *P* < .0001	43 (34-51)	36 (26-46)
Crizotinib (n = 138)	11.1 (9.1-13.0)		19 (12-27)	18 (11-26)

Final Analysis: INV-Assessed Systemic PFS



No. at Risk

Brigatinib	137	102	88	78	70	60	52	30	3
Crizotinib	138	80	46	35	22	18	17	7	1

INV-Assessed PFS	Median PFS (95% CI), months	HR (95% CI)	3-Year PFS Rate (95% CI), %	4-Year PFS Rate (95% CI), %
Brigatinib (n = 137)	30.8 (21.3-40.6)	0.43 (0.31-0.58) *P* < .0001	45 (36-54)	36 (26-46)
Crizotinib (n = 138)	9.2 (7.4-12.7)		18 (11-26)	16 (9-23)

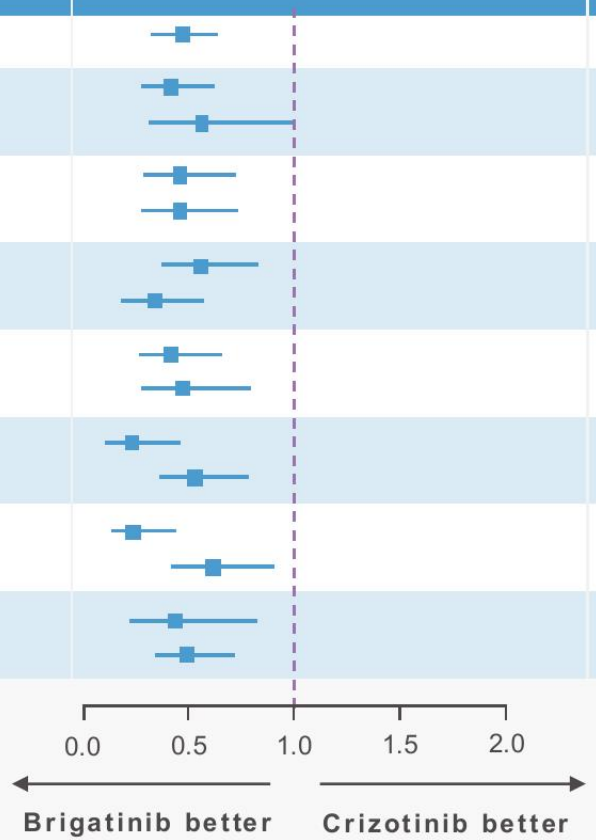
Final analysis data cutoff: January 29, 2021; median follow-up: 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm.

Camidge DR, et al. *J Thorac Oncol.* 2021;16:2091-2108.

ALTA-1L Final Analysis: BIRC-Assessed PFS by Subgroup

Final Analysis: BIRC-Assessed PFS by Subgroup

Subgroup		No. of Patients Brigatinib/ Crizotinib	HR for Disease Progression or Death (95% CI)
Overall		137/138	0.48 (0.35-0.66)
Age	18-64 years	93/95	0.42 (0.29-0.63)
	≥ 65 years	44/43	0.58 (0.33-1.01)
Sex	Female	69/81	0.47 (0.30-0.73)
	Male	68/57	0.48 (0.30-0.75)
Race	Non-Asian	78/89	0.56 (0.38-0.84)
	Asian	59/49	0.35 (0.20-0.59)
Smoking status[a]	Never smoker	84/75	0.43 (0.28-0.65)
	Former smoker	50/56	0.48 (0.29-0.80)
ECOG PS[b]	0	54/53	0.25 (0.13-0.48)
	1	76/78	0.54 (0.37-0.79)
Brain metastases at baseline[c]	Yes	40/41	0.25 (0.14-0.46)
	No	97/97	0.62 (0.43-0.91)
Prior chemotherapy[d]	Yes	36/37	0.45 (0.24-0.83)
	No	101/101	0.50 (0.35-0.73)



- Brigatinib had superior overall BIRC-assessed PFS vs crizotinib across almost all subgroups, most prominently in patients with baseline brain metastases and patients with ECOG PS of 0
- In patients with baseline brain metastases, brigatinib reduced the risk of BIRC-assessed progression or death by 75% vs crizotinib (HR, 0.25)

Final analysis data cutoff: January 29, 2021; median follow-up: 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm.

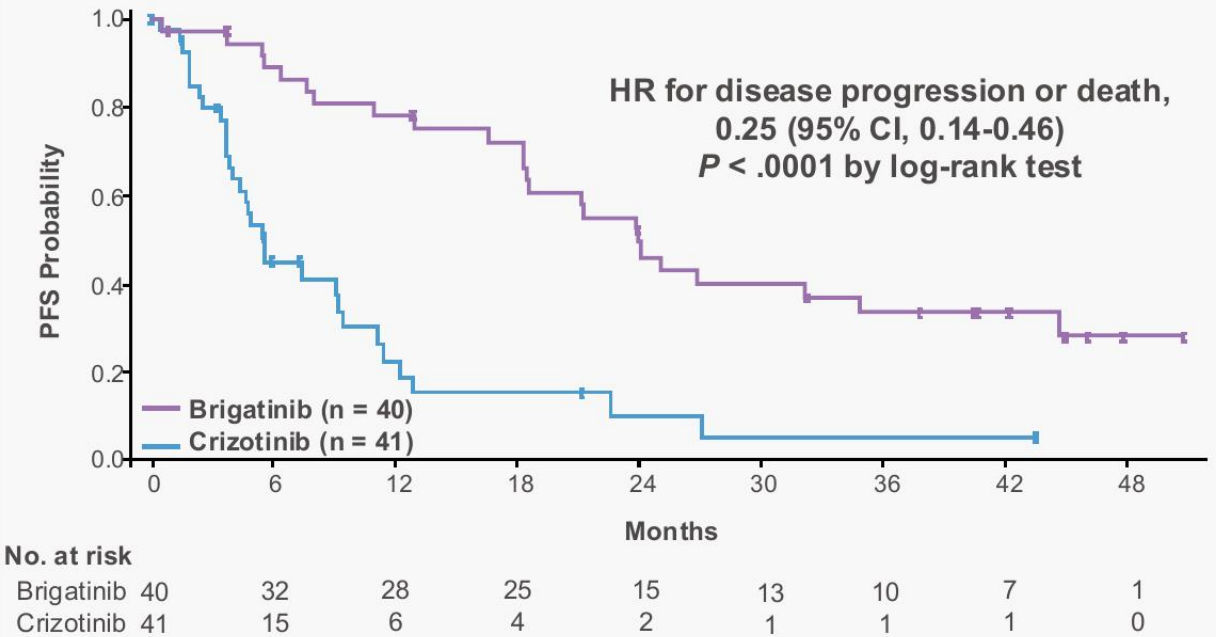
HRs were not calculated due to insufficient patient numbers for patients who were: [a] Current smokers (brigatinib, n = 3; crizotinib, n = 7) or [b] had ECOG PS of 2 (brigatinib, n = 7; crizotinib, n = 7); [c] Baseline brain metastases as assessed by INV; [d] Prior chemotherapy in a locally advanced or metastatic setting.

Camidge DR, et al. *J Thorac Oncol*. 2021;16:2091-2108.

ALTA-1L Final Analysis: BIRC-Assessed PFS by Brain Metastases Status at Baseline

- Patients with baseline brain metastases treated with brigatinib achieved a 75% reduction in the overall risk of progression or death vs crizotinib (BIRC assessed)

Final Analysis: Patients With Any Brain Metastases at Baseline

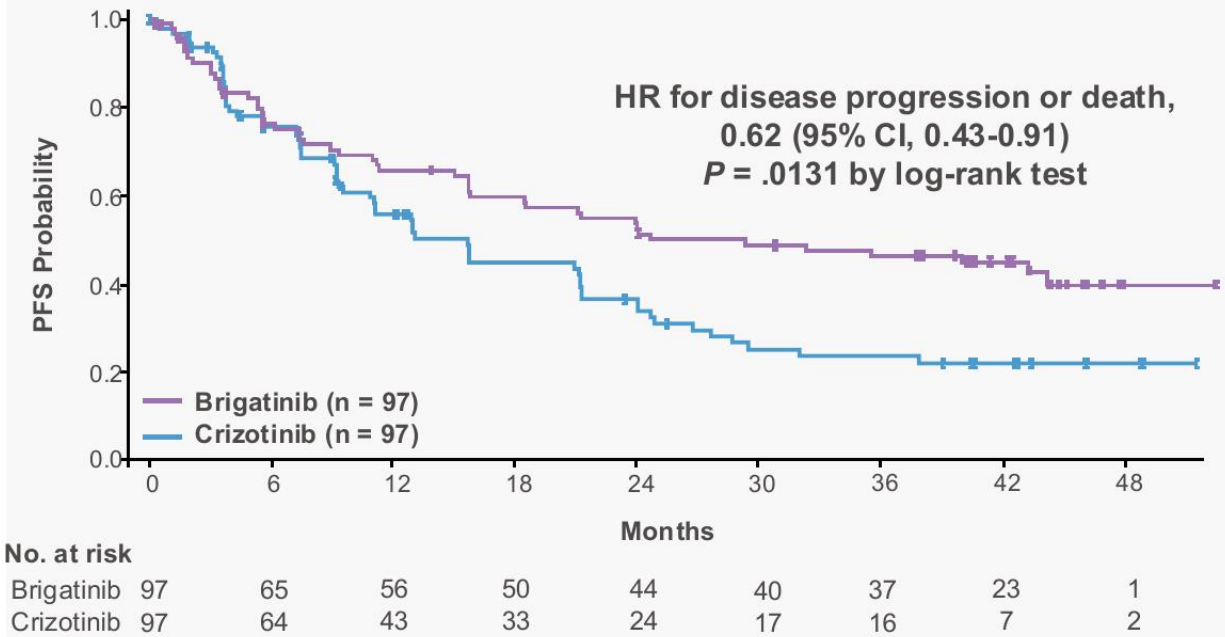


No. at risk

Brigatinib	40	32	28	25	15	13	10	7	1
Crizotinib	41	15	6	4	2	1	1	1	0

BIRC-Assessed PFS	Median PFS (95% CI), months	HR (95% CI)	3-Year PFS Rate (95% CI), %	4-Year PFS Rate (95% CI), %
Brigatinib (n = 40)[a]	24.0 (18.4-34.9)	0.25 (0.14-0.46) P < .0001	33 (18-49)	28 (13-45)
Crizotinib (n = 41)[a]	5.6 (3.8-9.2)		5 (0-20)	NE

Final Analysis: Patients Without Brain Metastases at Baseline



No. at risk

Brigatinib	97	65	56	50	44	40	37	23	1
Crizotinib	97	64	43	33	24	17	16	7	2

BIRC-Assessed PFS	Median PFS (95% CI), months	HR (95% CI)	3-Year PFS Rate (95% CI), %	4-Year PFS Rate (95% CI), %
Brigatinib (n = 97)[a]	29.3 (15.7-NE)	0.62 (0.43-0.91) P = .0131	47 (36-57)	40 (28-51)
Crizotinib (n = 97)[a]	15.6 (9.5-21.1)		24 (15-34)	22 (13-32)

Final analysis data cutoff: January 29, 2021; median follow-up: 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm.

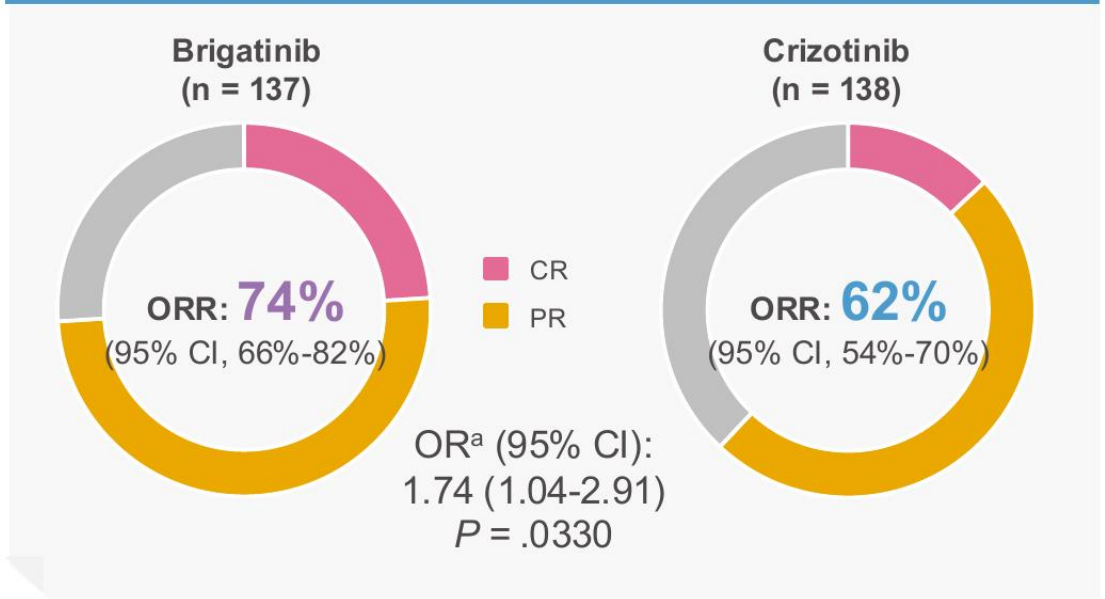
[a] Per INV assessment.

Company's data on file (CSR AP26113-13-301, Table 15.2.1.3.9, Figure 15.2.1.1.1.2, Figure 15.2.1.1.1.3).

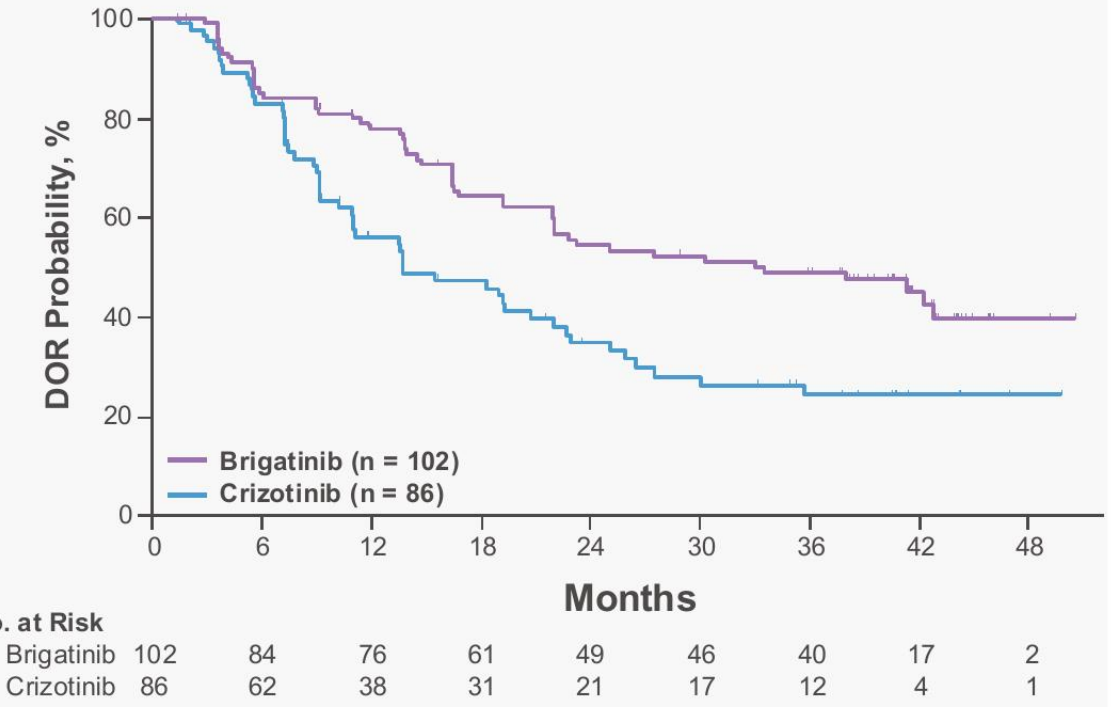
ALTA-1L Final Analysis: Overall Response and DOR (ITT population; BIRC assessed)

- Robust overall responses were observed with brigatinib treatment

Confirmed ORR



DOR in Confirmed Responders



Final Analysis: DOR in Confirmed Responders

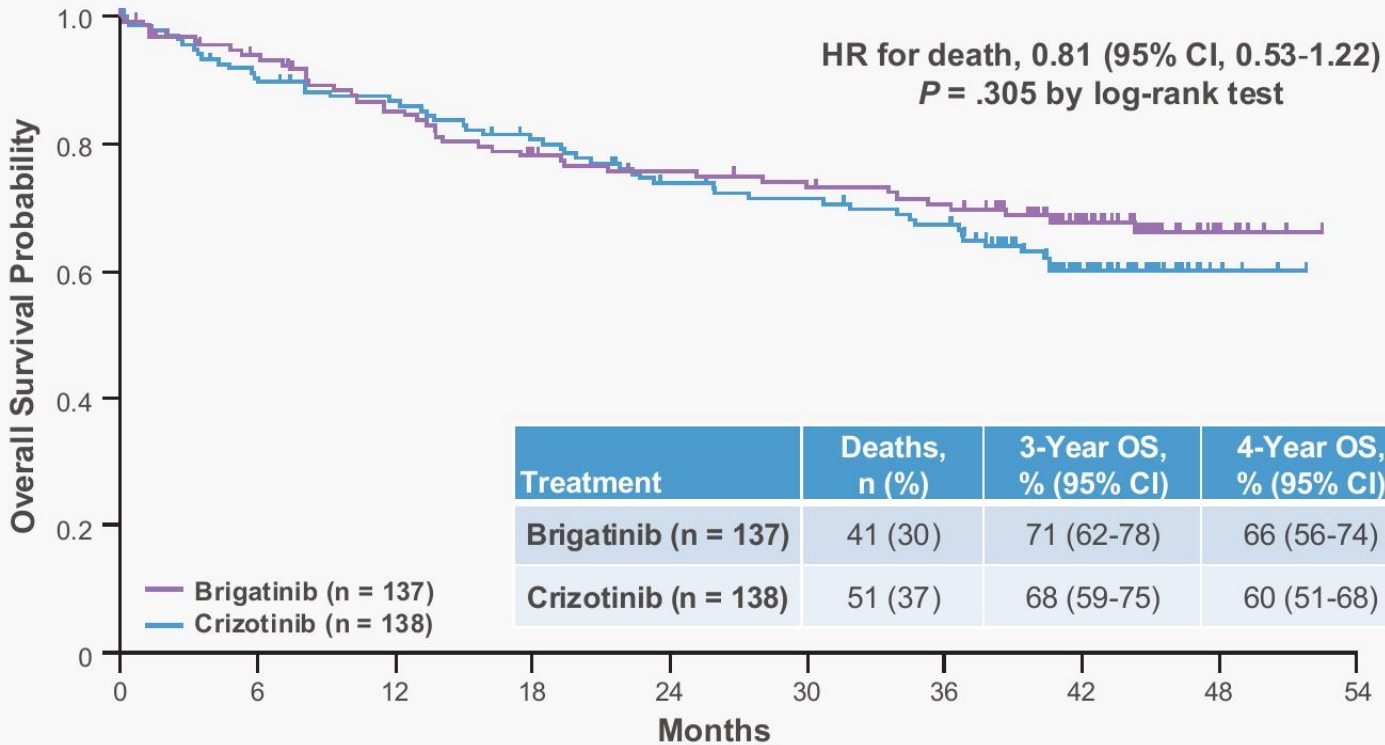
Treatment	Median DOR (95% CI), months	3-Year Probability[b] of Maintaining Response (95% CI), %	4-Year Probability[b] of Maintaining Response (95% CI), %
Brigatinib (n = 102)	**33.2** (22.1-NE)	49 (38-58)	40 (28-51)
Crizotinib (n = 86)	**13.8** (10.4-22.1)	25 (15-36)	25 (15-36)

Final analysis data cutoff: January 29, 2021; median follow-up: 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm.
[a] ORs (brigatinib vs crizotinib) and *P* values are from a Cochran-Mantel-Haenszel test stratified by presence of prior chemotherapy for locally advanced or metastatic disease; [b] Kaplan-Meier estimated probability of maintaining response.

Camidge DR, et al. *J Thorac Oncol*. 2021;16:2091-2108.

ALTA-1L Final Analysis: OS (ITT population)

OS



Treatment	Deaths, n (%)	3-Year OS, % (95% CI)	4-Year OS, % (95% CI)
Brigatinib (n = 137)	41 (30)	71 (62-78)	66 (56-74)
Crizotinib (n = 138)	51 (37)	68 (59-75)	60 (51-68)

No. at Risk

Brigatinib	137	121	108	97	92	88	84	52	7
Crizotinib	138	123	116	106	94	90	84	42	5

- OS data were still maturing at the final analysis (30% event rate in the brigatinib arm)
- 47% (65 patients) in the crizotinib arm crossed over to brigatinib therapy after PD

BIRC-Assessed Efficacy in Patients in the Crizotinib Arm Who Crossed Over to Brigatinib (n = 65)	
Median follow-up (range), months	22.7 (0.2-37.6)
Confirmed ORR (95% CI), %	57 (44-69)
Median DOR in confirmed responders (95% CI), months	19.1 (10.9-23.5)
Median PFS (95% CI), months	16.8 (10.1-23.9)

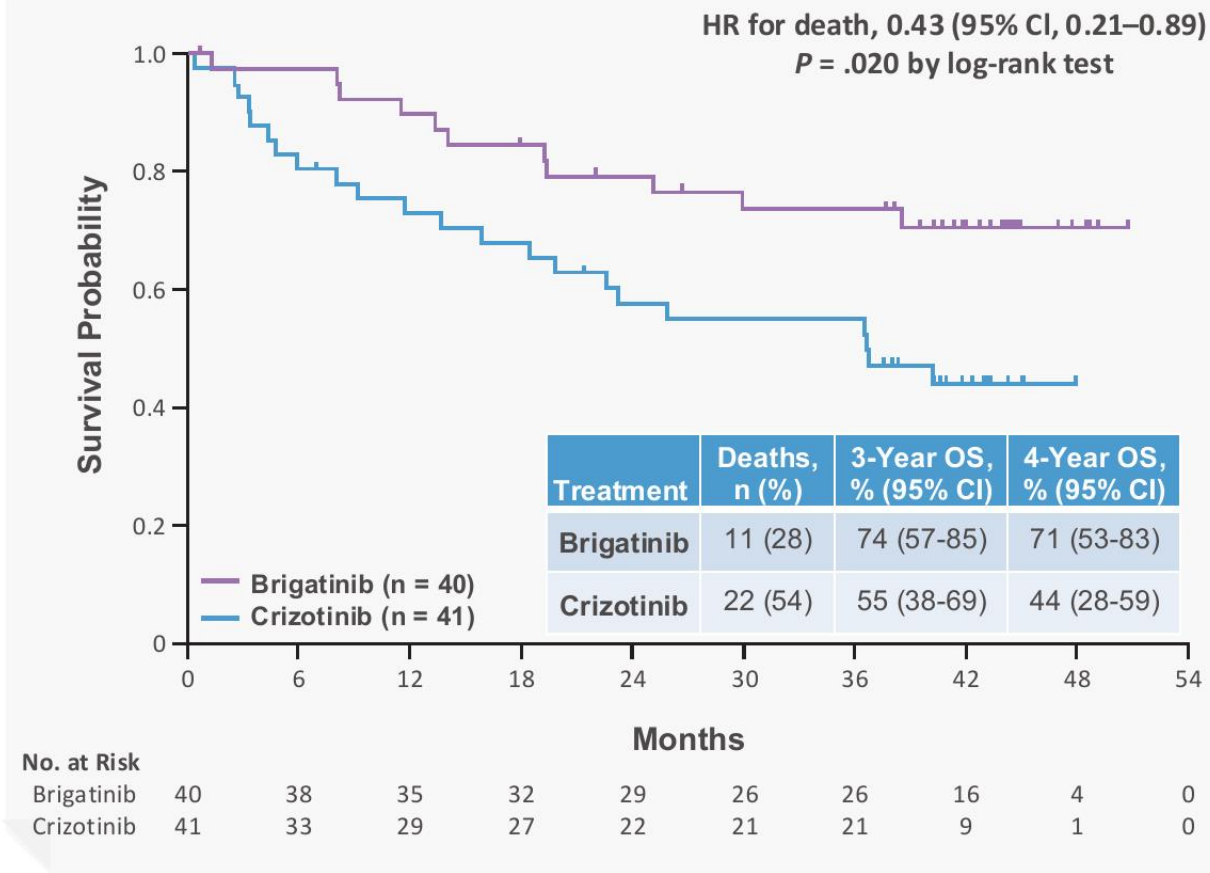
- In sensitivity analyses adjusted for treatment crossover in the crizotinib arm, the OS HR was 0.54 (95% CI, 0.31-0.92; *P* = .023) by the MSM method and 0.50 (95% CI, 0.28-0.87; *P* = .014) by the IPCW approach

Final analysis data cutoff: January 29, 2021; median follow-up: 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm.

Camidge DR, et al. *J Thorac Oncol*. 2021;16:2091-2108.

ALTA-1L Final Analysis: OS by Brain Metastases at Baseline (ITT population)

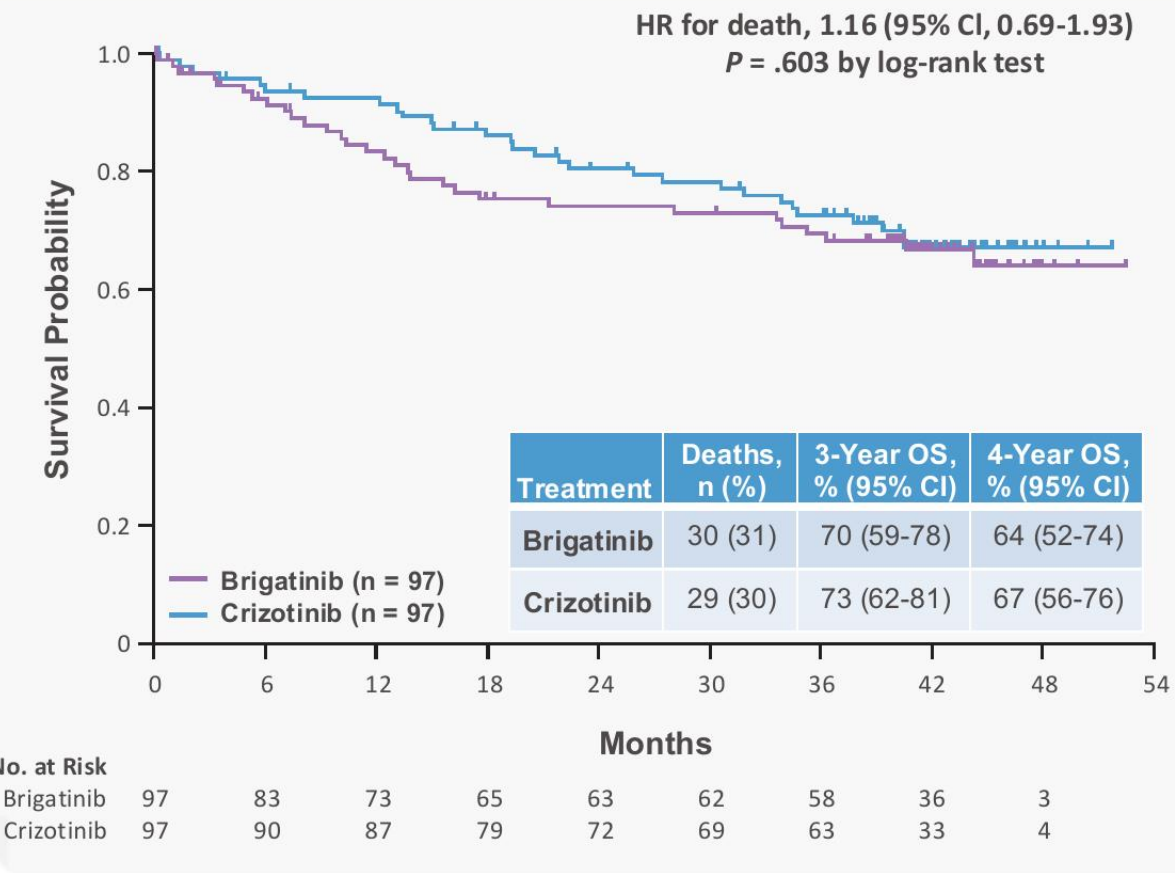
OS in Patients With Any Brain Metastases at Baseline[a]



Treatment	Deaths, n (%)	3-Year OS, % (95% CI)	4-Year OS, % (95% CI)
Brigatinib	11 (28)	74 (57-85)	71 (53-83)
Crizotinib	22 (54)	55 (38-69)	44 (28-59)

No. at Risk										
Brigatinib	40	38	35	32	29	26	26	16	4	0
Crizotinib	41	33	29	27	22	21	21	9	1	0

OS in Patients Without Brain Metastases at Baseline[a]



Treatment	Deaths, n (%)	3-Year OS, % (95% CI)	4-Year OS, % (95% CI)
Brigatinib	30 (31)	70 (59-78)	64 (52-74)
Crizotinib	29 (30)	73 (62-81)	67 (56-76)

No. at Risk										
Brigatinib	97	83	73	65	63	62	58	36	3	
Crizotinib	97	90	87	79	72	69	63	33	4	

Final analysis data cutoff: January 29, 2021; median follow-up: 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm.

[a] Brain metastasis present at baseline based on INV assessment.

Camidge DR, et al. *J Thorac Oncol.* 2021;16:2091-2108.

ALTA-1L: Safety profile* (1/2)

TEAE, %[1,2]	BRIGATINIB (n=136)		CRIZOTINIB (n=137)	
	ALL GRADE	GRADE ≥3	ALL GRADE	GRADE ≥3
Diarrhea	58	2	56	3
Increased blood CPK†	50	26	17	1
Cough	36	0	21	0
Nausea	33	2	59	3
Hypertension	32	14	9	4
Increased AST level	26	4	26	7
Back pain	26	1	17	2
Dyspnea	24	2	21	4
Headache	24	2	18	0
Increased lipase‡	24	15	17	8
Increased ALT level	23	4	36	10
Vomiting	22	1	45	2
Fatigue	21	0	23	1
Pruritis	21	1	6	1
Constipation	20	1	42	0
Arthralgia	20	1	12	0
Rash	19	0	3	0
Increased amylase‡	18	6	9	1

Final analysis data cut-off, Jan 29, 2021. Median follow-up, 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm
*Any grade reported in ≥10% or Grade ≥3 reported in >2% of patients; †Myalgia was reported in 14 (10%) and 11 (8%) patients in the brigatinib and crizotinib arms, respectively. Musculoskeletal pain was reported in 15 (11%) and 11 (8%) patients, respectively. No Grade ≥3 myalgia or musculoskeletal pain were reported in either arm; ‡No clinical cases of pancreatitis were reported in either arm
ALT, alanine aminotransferase; AST, aspartate aminotransferase; CPK, creatine phosphokinase; TEAE, treatment-emergent adverse event
1. Camidge DR, et al. J Thorac Oncol 2021;16:2091–108; 2. Takeda. Data on file (AP26113-13-301 Table 15.3.1.2.1.4)

ALTA-1L: Safety profile* (2/2)

TEAE, %[1,2]	BRIGATINIB (n=136)		CRIZOTINIB (n=137)	
	ALL GRADE	GRADE ≥3	ALL GRADE	GRADE ≥3
Decreased appetite	12	1	20	3
Increased blood alkaline phosphatase	13	3	13	1
Dyspepsia	11	0	17	1
Anemia	10	3	6	1
Pneumonia	10	5	7	4
Peripheral edema	10	1	46	1
Dermatitis acneiform	10	0	2	0
Increased blood cholesterol	10	1	1	0
Upper abdominal pain	8	1	19	2
Bradycardia	7	1	16	0
Pain in extremity	7	0	14	1
Increased blood creatine	6	0	15	0
Dysgeusia	4	0	15	0
Neoplasm progression	4	3	4	3
Pulmonary embolism†	2	2	6	4
Decreased neutrophil count	2	1	10	5
Photopsia	1	0	21	1
Visual impairment	1	0	17	0
GERD	1	0	12	0

Final analysis data cut-off, Jan 29, 2021. Median follow-up, 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm
*Any grade reported in ≥10% or Grade ≥3 reported in >2% of patients; †Any-grade and Grade ≥3 pulmonary embolism in the brigatinib arm occurred in 2% of patients
GERD, gastroesophageal reflux disease; TEAE, treatment-emergent adverse event
1. Camidge DR, et al. J Thorac Oncol 2021;16:2091–108; 2. Takeda. Data on file (AP26113-13-301 Table 15.3.1.2.1.4)

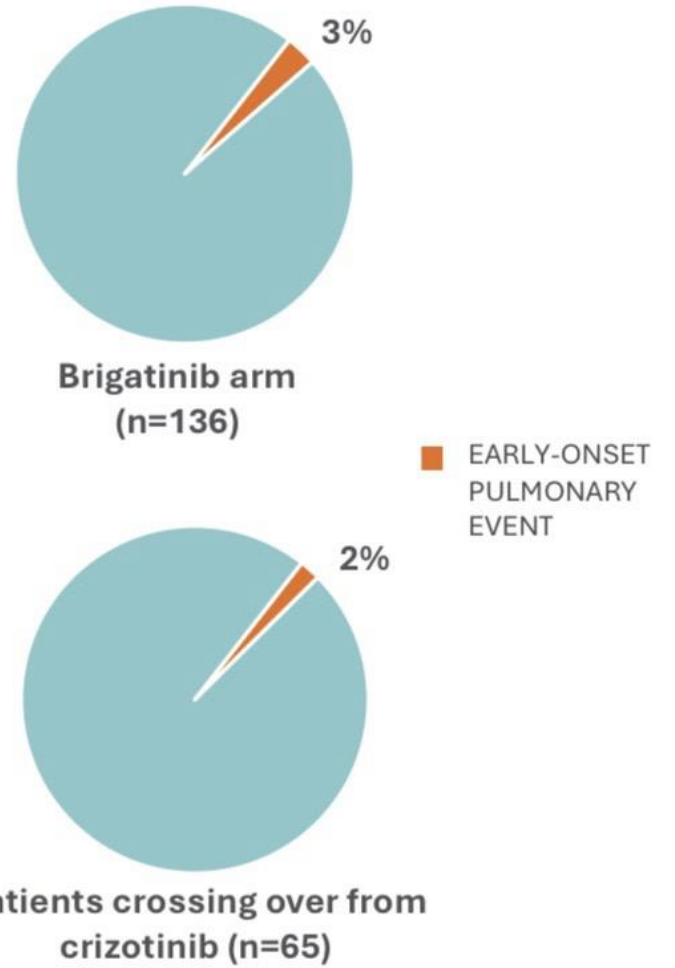
ALTA-1L: Early-onset pulmonary events

- **3% of patients (4/136) in the brigatinib arm had early-onset pulmonary events (within 14 days of treatment initiation)[1,2]**
- All 4 patients with early-onset pulmonary events discontinued brigatinib after the event, per protocol[1]
- Of 65 patients who crossed over from crizotinib to brigatinib, 1 (2%) experienced early-onset pulmonary events[2]

PULMONARY EVENTS IN THE ALTA-1L STUDY, n (%)[2,3]	BRIGATINIB (n=136)	CRIZOTINIB (n=137)	CRIZOTINIB → BRIGATINIB (n=65)
ILD/pneumonitis of any grade, occurring at any time[3]	8 (6)	3 (2)	4 (6)
Grade ≥3 ILD/pneumonitis at any time[3]	4 (3)	1 (<1)	1 (2)
ILD/pneumonitis of any grade, early onset[2]	4 (3)	0	1 (2)

- An observational study is ongoing to collect further information on early-onset pulmonary events with brigatinib (NCT03389399)

Early-onset pulmonary events in the ALTA-1L trial[2]

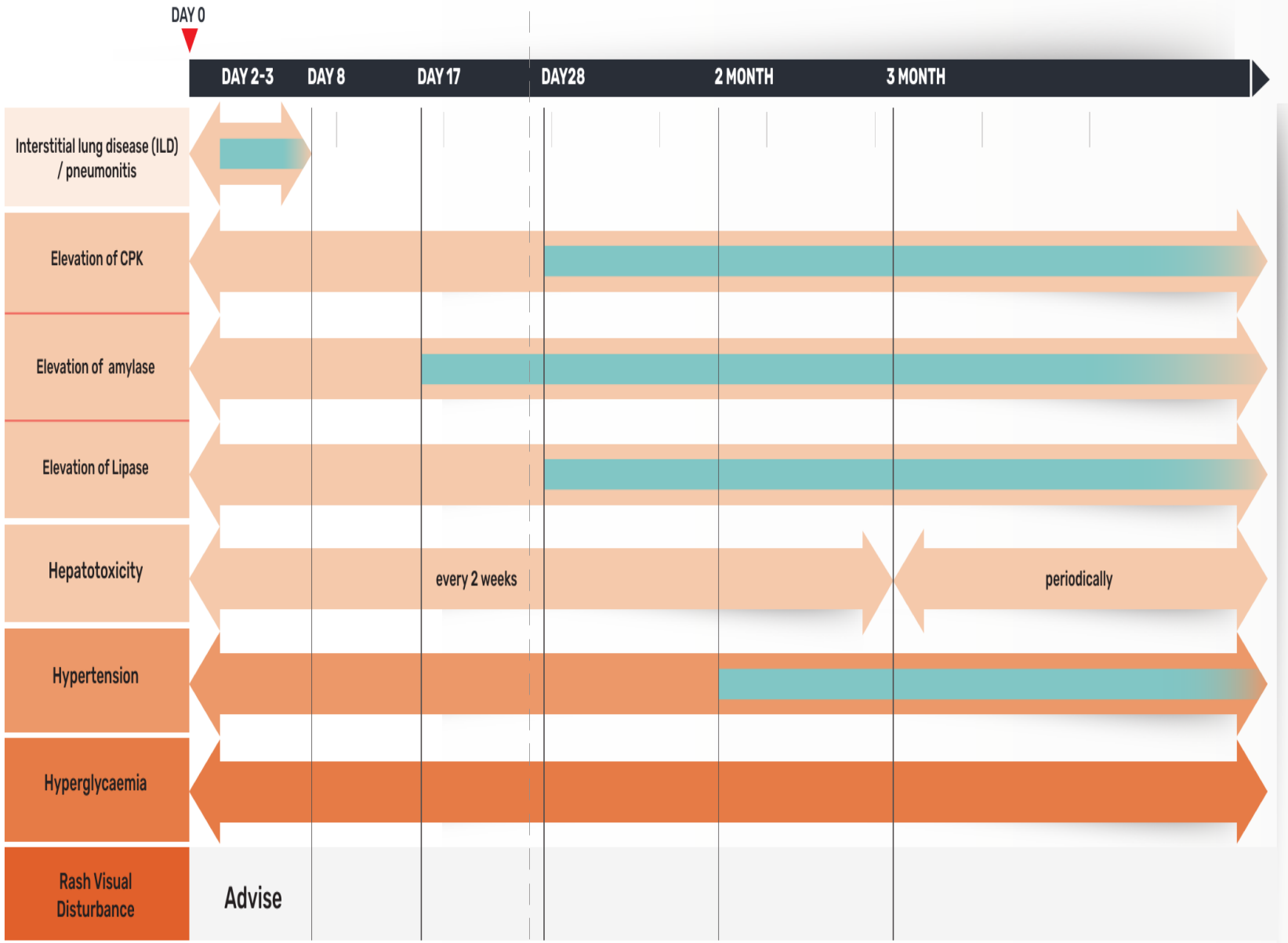


Final analysis data cut-off, Jan 29, 2021. Median follow-up, 40.4 months in the brigatinib arm and 15.2 months in the crizotinib arm
ILD, interstitial lung disease
1. Camidge DR, et al. N Engl J Med. 2018;379:2027–39; 2. Takeda. Data on file (CSR AP26113-13-301, Table 15.3.1.4.1, Table 15.3.1.4.1.1);
3. Camidge DR, et al. J Thorac Oncol 2021;16:2091–108

Comprehensive Patient Monitoring[1]

Recommended Evaluations:

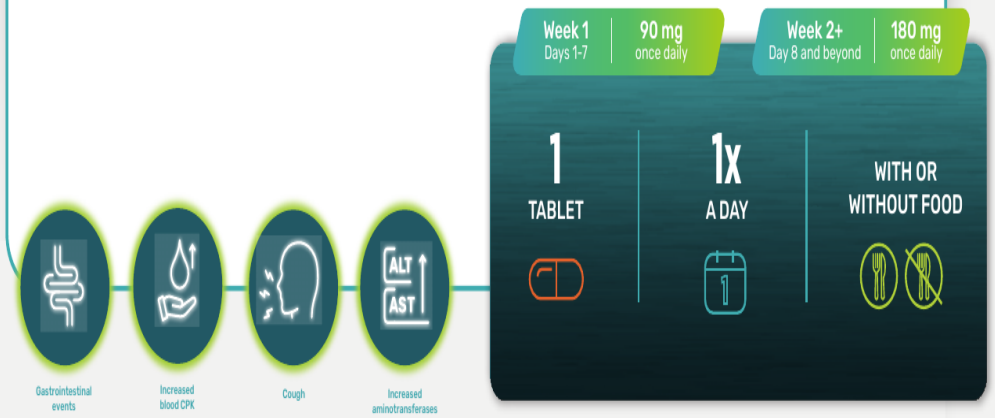
- Vital Signs Monitoring (including blood pressure, heart rate, and respiratory status)
- Complete Blood Count (CBC)
- Blood Chemistry Analysis (including liver and kidney function tests)
- Electrocardiogram (ECG) Assessments



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ALUNBRIG® has a well-established safety profile and long-term tolerability.[1,2,3]

- The most common (>25%) TEAEs of any grade included gastrointestinal events, increased blood CPK, cough, and increased aminotransferases
- The most common serious adverse reactions (≥2%) included [pneumonia, pneumonitis, dyspnoea, and pyrexia]
- Discontinuation rates due to AEs were comparable between treatment arms: 13% with ALUNBRIG vs 9% with crizotinib



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Recommended Dose Reduction Levels[1]

Dose	Dose reduction levels		
	First	Second	Third
90 mg once daily (first 7 days)	reduce to 60 mg once daily	permanently discontinue	not applicable
180 mg once daily	reduce to 120 mg once daily	reduce to 90 mg once daily	reduce to 60 mg once daily

*Brigatinib should be permanently discontinued if patient is unable to tolerate the 60 mg once daily dose

ILD/Pneumonitis[1,2,3]

Grade 1

- during the first 7 Day → • Withhold until recovery to BL • Resume at 90 mg dose • NOT escalate to 180 mg
- after the first 7 Day → • Withhold until recovery to BL • Resume at 180 mg dose
- recur → Discontinued

Grade 2

- during the first 7 Day → • Withhold until recovery to BL • Resume at 30 mg + 30 mg dose • NOT escalate to 180 mg
- after the first 7 Day → • Withhold until recovery to BL • Resume at 30 mg + 90 mg dose
- recur → Discontinued

Grade 3/4 → Discontinued

CTCAE Pulmonary Event[1,2,3]

Grade					
Adverse Event	1	2	3	4	5
Pneumonitis	Asymptomatic; clinical or diagnostic observations only; intervention not indicated	Symptomatic; medical intervention indicated; limiting instrumental ADL	Severe symptoms; limiting self care ADL; oxygen indicated	Life-threatening respiratory compromise; urgent intervention indicated (e.g., tracheotomy or intubation)	Death

Definition: A disorder characterized by inflammation focally or diffusely affecting the lung parenchyma.

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Hypertension[1,2,3]

Grade 3

- First Episode → • Withhold until recovery to grade ≤ 1 (SBP <140 mmHg and DBP<90mmHg) • Resume at 90 mg dose
- recur → • Withhold until recovery to grade ≤ 1 • Resume at NEXT LOWER dose or discontinued

- SBP ≥160 mm Hg or DBP ≥100 mmHg medical intervention indicated
- ≥1 antihypertensive drug or more intensive therapy than previously used indicated

Grade 4

Life threatening consequences, urgent intervention indicated

- First Episode → • Withhold until recovery to grade ≤ 1 (SBP <140 mmHg and DBP<90mmHg) • Resume at 60 mg dose or discontinue
- recur → Discontinued

CPK Elevation[1,2,3]

Grade 3 (>5.0 x ULN)

- 1st Episode → • Withhold until recovery to Gr1/ BL • Resume at SAME dose
- Recurrent → • Withhold until recovery to BL • Resume at NEXT LOWER dose

Grade 4 (>10.0 x ULN) → • Withhold until recovery to Gr1/ BL • Resume at NEXT LOWER dose

Pancreatic Enzyme Elevation[1,2,3]

Grade 3 lipase/ amylase elevation (>2.0 x ULN)

- 1st Episode
  - Withhold until recovery to Gr1/ BL
  - Resume at **SAME** dose
- Recurrent
  - Withhold until recovery to BL
  - Resume at **NEXT LOWER** dose

Grade 4 lipase/ amylase elevation (>5.0 x ULN)

- Withhold until recovery to Gr1/ BL
- Resume at **NEXT LOWER** dose

Hepatotoxicity[1,2,3]

Grade ≥ 2 (>3 x ULN) elevation of ALT or AST with concurrent total bilirubin elevation >2 x ULN in the absence of cholestasis or haemolysis

- Permanently discontinued

Grade ≥ 3 (>5 x ULN) elevation of either ALT or AST with bilirubin ≤2 x ULN

- Withhold until recovery to BL/ ≤3 x ULN
- Resume at **NEXT LOWER** dose

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Other Adverse Event

Grade 3

- First Episode
  - Brigatinib should be withheld until recovery to baseline
  - Resume at **SAME** dose
- If Grade 3 recur
  - Withhold until recovery to baseline
  - Resume at **NEXT LOWER** dose or discontinued

Grade 4

- First Episode
  - Brigatinib should be withheld until recovery to baseline
  - Resume at **NEXT LOWER** dose or discontinue
- If Grade 4 recur
  - Brigatinib should be withheld until recovery to baseline
  - Resume at **NEXT LOWER** dose or discontinue

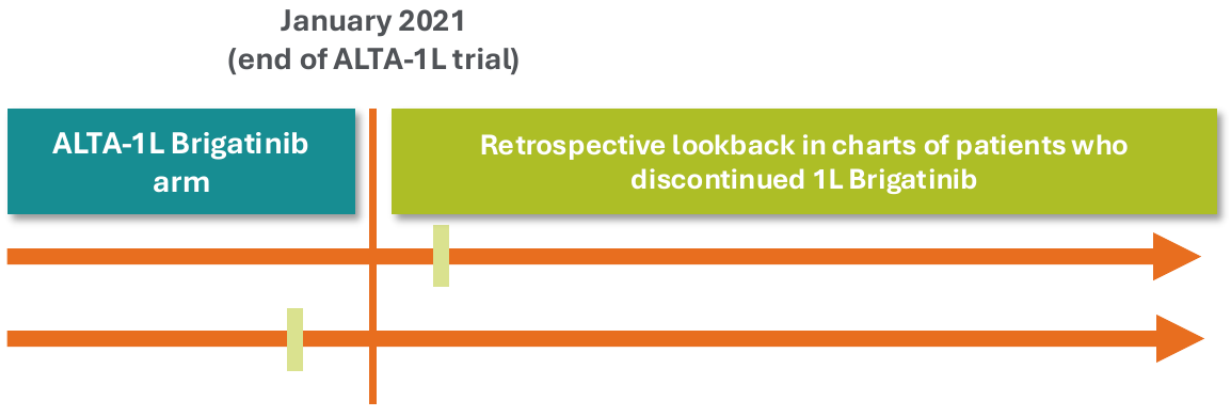
ALTA-1L retrospective chart review

(Brigatinib-5010)



Objectives

A retrospective, longitudinal, non-interventional, multinational, multisite-based chart review study to investigate the real-world treatment patterns and outcomes of subsequent treatments following discontinuation of 1L Brigatinib in the ALTA-1L trial



- Patients were followed from the index date[a] until end of follow-up or death
- Data were extracted from patient medical records (electronic case report forms), between November 4 2022 and October 18 2023 (database lock)

Study outcomes

- **rwTTD:** time from initiation to discontinuation of 2L treatment for any reason
- **rwPFS:** time elapsed from initiation of 2L treatment to disease progression or death
- **rwPFS2:** time from randomization in ALTA-1L trial to first documented disease progression on 2L treatment or death due to any cause among those patients who initiated subsequent systemic anticancer treatments
- **OS:** was defined as time from the date of randomization in the ALTA-1L trial until date of death

[a]Defined as date of final dose of 1L brigatinib

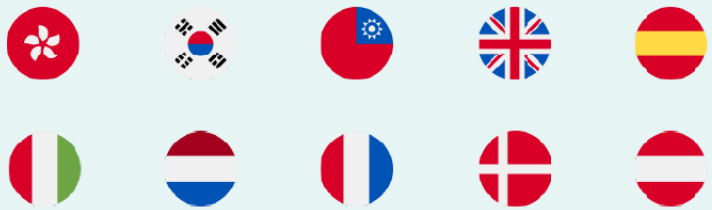
1L, first line; 2L, second line; OS, overall survival; rwPFS, real-world progression-free survival; rwPFS2, real-world PFS2; rwTTD, real-world time to treatment discontinuation

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Baseline patient characteristics

Patients

- 20 sites enrolled patients across 10 countries



- 48 patients met the inclusion criteria and were enrolled; median follow-up was 12.4 months
- The majority of patients were female, identified as Asian and were never smokers
- The most common reason for 1L Brigatinib discontinuation was PD (n= 34 [70.9%])
- Median time from index date to start of 2L (95% CI) was 4.0 (3.0–9.0) days

	Enrolled set (n=48)	2L ALK TKI (n=30)	2L lorlatinib (n=16)
Age (years), median (range)	58 (33–85)	57.5 (34–82)	52.5 (40–77)
Female sex, n (%)	26 (54.2)	18 (60.0)	10 (62.5)
Race, n (%)			
Native Asian	26 (54.2)	15 (50.0)	7 (43.8)
White	21 (43.8)	14 (46.7)	8 (50.0)
Unknown	1 (2.1)	1 (3.3)	1 (6.3)
Geographic region, n (%)			
Asia Pacific	25 (52.1)	15 (50.0)	7 (43.8)
Europe	23 (47.9)	15 (50.0)	9 (56.3)
Cigarette smoking, n (%)			
Never	30 (62.5)	20 (66.7)	12 (75.0)
Current	2 (4.2)	0	0
Former	13 (27.1)	8 (26.7)	3 (18.8)
Unknown	3 (6.3)	2 (6.7)	1 (6.3)
Brain metastases, n (%)	20 (41.7)	16 (53.3)	10 (62.5)
Follow-up (months) median (IQR)	12.4 (3.7, 27.0)	17.0 (6.1, 36.0)	14.8 (2.7, 33.2)

1L, first line; 2L, second line; ALK, anaplastic lymphoma kinase; CI, confidence interval; IQR, interquartile range; PD, progressive disease; TKI, tyrosine kinase inhibitor

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Real-world PFS, PFS2 and OS

rwPFS

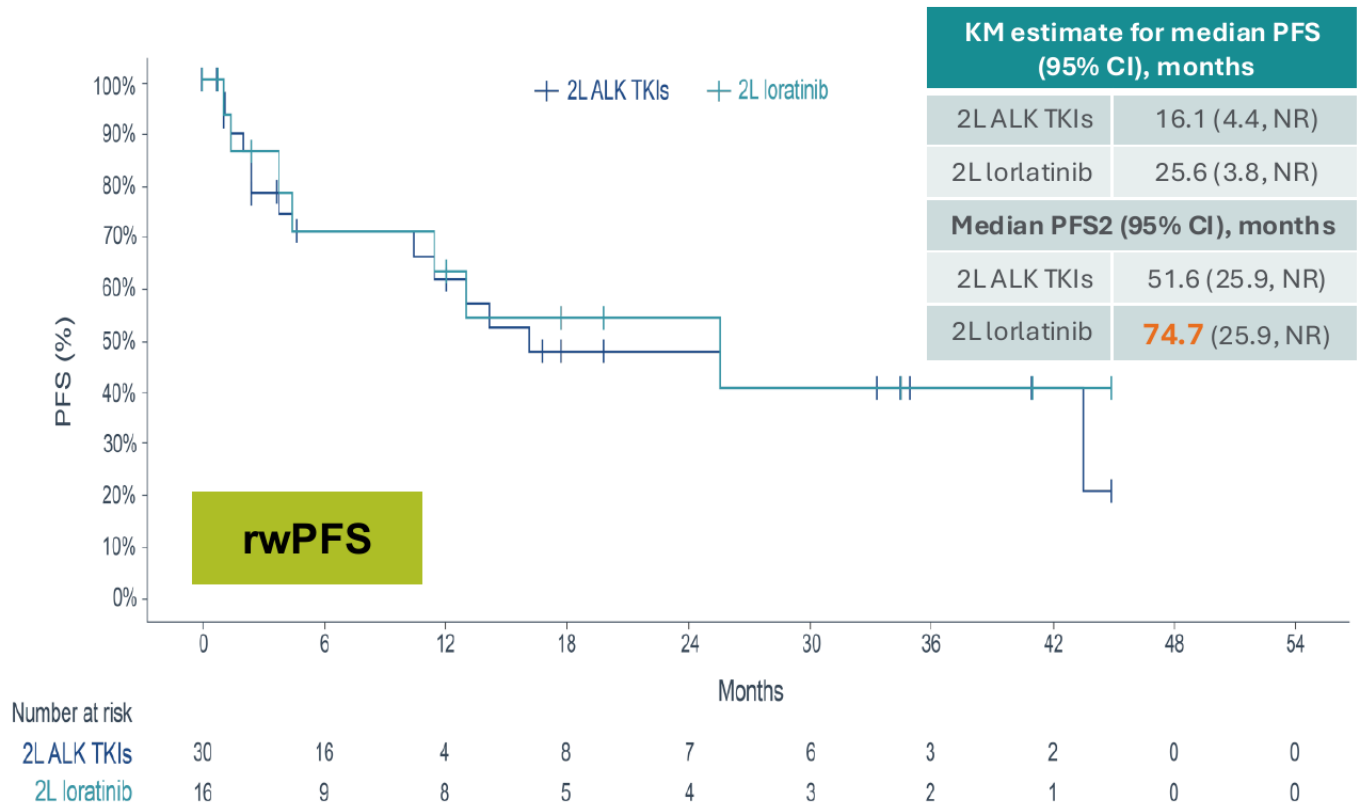
- Estimated 24-month rwPFS (95% CI) for 2L ALK TKIs: 47.0% (26.2, 65.3)
- Estimated 24-month rwPFS (95% CI) for 2L Lorlatinib: 53.4% (23.9, 76.0)

rwPFS2

- Estimated 24-month rwPFS2 (95% CI) for 2L ALK TKIs: 78.4% (58.1, 89.7)
- Estimated 24-month rwPFS2 (95% CI) for 2L Lorlatinib: 86.7% (56.4, 96.5)

rwOS

- mOS (95% CI): **74.7** (30.0, NR) months for both 2L ALK TKIs and 2L lorlatinib
- Estimated 36-month OS (95% CI): 66.7% (46.9, 80.5) for 2L ALK TKIs and 75.0% (46.3, 89.8) for 2L lorlatinib



KM estimate for median PFS (95% CI), months	
2L ALK TKIs	16.1 (4.4, NR)
2L lorlatinib	25.6 (3.8, NR)
Median PFS2 (95% CI), months	
2L ALK TKIs	51.6 (25.9, NR)
2L lorlatinib	**74.7** (25.9, NR)

2L, second-line; ALK, anaplastic lymphoma kinase; CI, confidence interval; KM, Kaplan Meier; mOS, median overall survival; NR, not reached; OS, overall survival; PFS, progression-free survival; rwOS, real world overall survival; rwPFS, real-world progression-free survival; rwPFS2, real-world PFS2; TKI, tyrosine kinase inhibitors; TTD, time-to-treatment discontinuation

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THANK YOU

